

SIMATS ENGINEERING

ASSESSMENT TOOL 3 - ANSWER KEY

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

Annexure A - Debugging & Optimisation Answers

Answer 1: Workstation Configuration Debugging (192.168.10.65 /26)

Configured: IP 192.168.10.65, Subnet Mask 255.255.255.192 (/26), Gateway 192.168.10.1. Target subnet: 192.168.10.0/26.

1. Subnet to Which the Configured IP Belongs

192.168.10.65 with a /26 mask (block size 64) actually falls in the range 192.168.10.64 - 192.168.10.127, i.e. it belongs to subnet 192.168.10.64/26 - NOT to 192.168.10.0/26 as intended.

2. Network Address, Broadcast Address and Valid Host Range (Intended Subnet 192.168.10.0/26)

Subnet	Network Address	Broadcast Address	Valid Host Range
192.168.10.0/26 (intended)	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
192.168.10.64/26 (actual, wrong)	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Validity Check of Configured IP and Gateway

- IP 192.168.10.65: INVALID for the intended subnet 192.168.10.0/26 - it actually lies in the neighboring subnet 192.168.10.64/26.
- Gateway 192.168.10.1: VALID for subnet 192.168.10.0/26 (it is the first usable host address of that subnet).

4. Configuration Errors Identified

- The workstation's IP address (192.168.10.65) and its gateway (192.168.10.1) lie in two different subnets.
- Because the host and gateway are not on the same subnet, the workstation cannot reach the gateway or any device in 192.168.10.0/26.

5. Optimized (Corrected) Configuration

- Assign an IP address within the valid range of 192.168.10.0/26, e.g. 192.168.10.2 (avoiding .1, which is reserved for the gateway).
- Subnet Mask: 255.255.255.192 (/26) - unchanged, it was already correct.
- Default Gateway: 192.168.10.1 - unchanged, keep as is.

6. Total Usable Host Addresses After Correction

For a /26 subnet: $2^6 - 2 = 62$ usable host addresses.

Answer 2: IPv6 Configuration Debugging (Device A & Device B)

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 3: Optimizing VLSM Subnet Allocation (192.168.1.0/24)

The /24 block was subnetted using /26, /27 and /28 masks. Using the addresses in the order allocated:

1 & 2. Network Address, Broadcast Address, Host Range, Usable & Unused Addresses

Subnet (as allocated)	Network Addr.	Broadcast Addr.	Valid Host Range	Usable Hosts
192.168.1.0/26	192.168.1.0	192.168.1.63	192.168.1.1 - 192.168.1.62	62
192.168.1.64/27	192.168.1.64	192.168.1.95	192.168.1.65 - 192.168.1.94	30
192.168.1.96/28	192.168.1.96	192.168.1.111	192.168.1.97 - 192.168.1.110	14

Unused block remaining after these three allocations: 192.168.1.112 - 192.168.1.255 (144 addresses, over half of the original /24 left unassigned).

3. Inefficient Subnet Allocations Identified

- Fixed subnet sizes (/26, /27, /28) were assigned without matching actual department host counts, so some departments run out of addresses while a large unused block (144 addresses) remains idle.
- A department needing more than 30 or 14 hosts, but placed in a /27 or /28 subnet, will face address shortage even though free space exists elsewhere in the /24 block.

4. Debugging the Addressing Plan

The root cause is that subnet sizes were chosen arbitrarily instead of being derived from actual/projected host requirements per department (classic error in non-VLSM-aware planning).

5. Recommended Optimized VLSM Scheme

Re-allocate subnets strictly according to actual host requirements, largest first, to minimize wastage. Example VLSM redesign:

Department (example need)	Subnet Assigned	Usable Hosts	Address Range
Dept 1 (~50 hosts)	192.168.1.0/26	62	192.168.1.1 - 192.168.1.62
Dept 2 (~25 hosts)	192.168.1.64/27	30	192.168.1.65 - 192.168.1.94
Dept 3 (~10 hosts)	192.168.1.96/28	14	192.168.1.97 - 192.168.1.110
Dept 4 (~5 hosts)	192.168.1.112/29	6	192.168.1.113 - 192.168.1.118

Remaining block 192.168.1.120 - 192.168.1.255 (136 addresses) is reserved for future departmental growth, giving a much more efficient utilization of the /24 block than the original fixed /26-/27-/28 plan.

Answer 4: IPv6 Configuration Debugging (Duplicate of Question 2)

Question 4 repeats Question 2 verbatim (same Device A and Device B addresses); the analysis and answers are identical to Answer 2 above.

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 5: Routing Table Debugging

Routing table: 192.168.1.0/24 -> 10.0.0.2, 192.168.2.0/24 -> 10.0.0.3, Default Route -> 10.0.0.1.

Reported issue: packets destined for 192.168.2.100 are not reaching the intended network.

1. Matching Routing Entry (Longest Prefix Match)

192.168.2.100 falls within 192.168.2.0 - 192.168.2.255, so it matches the entry 192.168.2.0/24 (the only and therefore longest matching prefix).

2. Next-Hop Router Selected

Next hop = 10.0.0.3, as per the matching routing table entry for 192.168.2.0/24.

3. Verification of Destination Subnet

192.168.2.100 is a valid host address within 192.168.2.0/24 (valid host range 192.168.2.1 - 192.168.2.254), so the destination address itself is correctly within the intended subnet.

4. Routing Configuration Errors Identified

The routing table entry itself is correct (192.168.2.0/24 via 10.0.0.3), so the fault is NOT a missing or wrong route entry. The likely causes of failure are operational, such as:

- Next-hop 10.0.0.3 is unreachable, down, or has no return path back to the source network
- Missing reciprocal/return route on the 10.0.0.3 router pointing back to the source subnet
- Interface connected to 10.0.0.3 is administratively down or misconfigured
- An ACL or firewall rule is silently dropping packets destined for 192.168.2.0/24

5. Optimizing the Routing Table

- Verify Layer-2/Layer-3 reachability of next-hop 10.0.0.3 (ping/traceroute) and correct any interface/link fault.
- Add a floating static route or redundant path as backup in case the primary next hop fails.
- Where the topology allows, replace static routes with a dynamic routing protocol (e.g. OSPF) so failed paths are detected and rerouted automatically.
- Confirm the return route exists on the 10.0.0.3 router so response packets are not dropped.

6. Route Used When No Matching Destination Exists

If a destination does not match 192.168.1.0/24 or 192.168.2.0/24, the router forwards it via the Default Route, next hop 10.0.0.1.

Answer 1: Workstation Configuration Debugging (192.168.10.65 /26)

Configured: IP 192.168.10.65, Subnet Mask 255.255.255.192 (/26), Gateway 192.168.10.1. Target subnet: 192.168.10.0/26.

1. Subnet to Which the Configured IP Belongs

192.168.10.65 with a /26 mask (block size 64) actually falls in the range 192.168.10.64 - 192.168.10.127, i.e. it belongs to subnet 192.168.10.64/26 - NOT to 192.168.10.0/26 as intended.

2. Network Address, Broadcast Address and Valid Host Range (Intended Subnet 192.168.10.0/26)

Subnet	Network Address	Broadcast Address	Valid Host Range
192.168.10.0/26 (intended)	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
192.168.10.64/26 (actual, wrong)	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Validity Check of Configured IP and Gateway

- IP 192.168.10.65: INVALID for the intended subnet 192.168.10.0/26 - it actually lies in the neighboring subnet 192.168.10.64/26.
- Gateway 192.168.10.1: VALID for subnet 192.168.10.0/26 (it is the first usable host address of that subnet).

4. Configuration Errors Identified

- The workstation's IP address (192.168.10.65) and its gateway (192.168.10.1) lie in two different subnets.
- Because the host and gateway are not on the same subnet, the workstation cannot reach the gateway or any device in 192.168.10.0/26.

5. Optimized (Corrected) Configuration

- Assign an IP address within the valid range of 192.168.10.0/26, e.g. 192.168.10.2 (avoiding .1, which is reserved for the gateway).
- Subnet Mask: 255.255.255.192 (/26) - unchanged, it was already correct.
- Default Gateway: 192.168.10.1 - unchanged, keep as is.

6. Total Usable Host Addresses After Correction

For a /26 subnet: $2^6 - 2 = 62$ usable host addresses.

Answer 2: IPv6 Configuration Debugging (Device A & Device B)

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
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- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
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Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 3: Optimizing VLSM Subnet Allocation (192.168.1.0/24)

The /24 block was subnetted using /26, /27 and /28 masks. Using the addresses in the order allocated:

1 & 2. Network Address, Broadcast Address, Host Range, Usable & Unused Addresses

Subnet (as allocated)	Network Addr.	Broadcast Addr.	Valid Host Range	Usable Hosts
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Unused block remaining after these three allocations: 192.168.1.112 - 192.168.1.255 (144 addresses, over half of the original /24 left unassigned).

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The root cause is that subnet sizes were chosen arbitrarily instead of being derived from actual/projected host requirements per department (classic error in non-VLSM-aware planning).

5. Recommended Optimized VLSM Scheme

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Dept 4 (~5 hosts)	192.168.1.112/29	6	192.168.1.113 - 192.168.1.118

Remaining block 192.168.1.120 - 192.168.1.255 (136 addresses) is reserved for future departmental growth, giving a much more efficient utilization of the /24 block than the original fixed /26-/27-/28 plan.

Answer 4: IPv6 Configuration Debugging (Duplicate of Question 2)

Question 4 repeats Question 2 verbatim (same Device A and Device B addresses); the analysis and answers are identical to Answer 2 above.

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329

- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
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3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

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There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

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Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
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Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 5: Routing Table Debugging

Routing table: 192.168.1.0/24 -> 10.0.0.2, 192.168.2.0/24 -> 10.0.0.3, Default Route -> 10.0.0.1.

Reported issue: packets destined for 192.168.2.100 are not reaching the intended network.

1. Matching Routing Entry (Longest Prefix Match)

192.168.2.100 falls within 192.168.2.0 - 192.168.2.255, so it matches the entry 192.168.2.0/24 (the only and therefore longest matching prefix).

2. Next-Hop Router Selected

Next hop = 10.0.0.3, as per the matching routing table entry for 192.168.2.0/24.

3. Verification of Destination Subnet

192.168.2.100 is a valid host address within 192.168.2.0/24 (valid host range 192.168.2.1 - 192.168.2.254), so the destination address itself is correctly within the intended subnet.

4. Routing Configuration Errors Identified

The routing table entry itself is correct (192.168.2.0/24 via 10.0.0.3), so the fault is NOT a missing or wrong route entry. The likely causes of failure are operational, such as:

- Next-hop 10.0.0.3 is unreachable, down, or has no return path back to the source network
- Missing reciprocal/return route on the 10.0.0.3 router pointing back to the source subnet
- Interface connected to 10.0.0.3 is administratively down or misconfigured
- An ACL or firewall rule is silently dropping packets destined for 192.168.2.0/24

5. Optimizing the Routing Table

- Verify Layer-2/Layer-3 reachability of next-hop 10.0.0.3 (ping/traceroute) and correct any interface/link fault.
- Add a floating static route or redundant path as backup in case the primary next hop fails.
- Where the topology allows, replace static routes with a dynamic routing protocol (e.g. OSPF) so failed paths are detected and rerouted automatically.
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- Assign an IP address within the valid range of 192.168.10.0/26, e.g. 192.168.10.2 (avoiding .1, which is reserved for the gateway).
- Subnet Mask: 255.255.255.192 (/26) - unchanged, it was already correct.
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6. Total Usable Host Addresses After Correction

For a /26 subnet: $2^6 - 2 = 62$ usable host addresses.

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1. Expanded (Complete) Notation

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Dept 4 (~5 hosts)	192.168.1.112/29	6	192.168.1.113 - 192.168.1.118

Remaining block 192.168.1.120 - 192.168.1.255 (136 addresses) is reserved for future departmental growth, giving a much more efficient utilization of the /24 block than the original fixed /26-/27-/28 plan.

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Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64

- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 5: Routing Table Debugging

Routing table: 192.168.1.0/24 -> 10.0.0.2, 192.168.2.0/24 -> 10.0.0.3, Default Route -> 10.0.0.1.

Reported issue: packets destined for 192.168.2.100 are not reaching the intended network.

1. Matching Routing Entry (Longest Prefix Match)

192.168.2.100 falls within 192.168.2.0 - 192.168.2.255, so it matches the entry 192.168.2.0/24 (the only and therefore longest matching prefix).

2. Next-Hop Router Selected

Next hop = 10.0.0.3, as per the matching routing table entry for 192.168.2.0/24.

3. Verification of Destination Subnet

192.168.2.100 is a valid host address within 192.168.2.0/24 (valid host range 192.168.2.1 - 192.168.2.254), so the destination address itself is correctly within the intended subnet.

4. Routing Configuration Errors Identified

The routing table entry itself is correct (192.168.2.0/24 via 10.0.0.3), so the fault is NOT a missing or wrong route entry. The likely causes of failure are operational, such as:

- Next-hop 10.0.0.3 is unreachable, down, or has no return path back to the source network
- Missing reciprocal/return route on the 10.0.0.3 router pointing back to the source subnet
- Interface connected to 10.0.0.3 is administratively down or misconfigured
- An ACL or firewall rule is silently dropping packets destined for 192.168.2.0/24

5. Optimizing the Routing Table

- Verify Layer-2/Layer-3 reachability of next-hop 10.0.0.3 (ping/traceroute) and correct any interface/link fault.
- Add a floating static route or redundant path as backup in case the primary next hop fails.
- Where the topology allows, replace static routes with a dynamic routing protocol (e.g. OSPF) so failed paths are detected and rerouted automatically.
- Confirm the return route exists on the 10.0.0.3 router so response packets are not dropped.

6. Route Used When No Matching Destination Exists

If a destination does not match 192.168.1.0/24 or 192.168.2.0/24, the router forwards it via the Default Route, next hop 10.0.0.1.

Answer 1: Workstation Configuration Debugging (192.168.10.65 /26)

Configured: IP 192.168.10.65, Subnet Mask 255.255.255.192 (/26), Gateway 192.168.10.1. Target subnet: 192.168.10.0/26.

1. Subnet to Which the Configured IP Belongs

192.168.10.65 with a /26 mask (block size 64) actually falls in the range 192.168.10.64 - 192.168.10.127, i.e. it belongs to subnet 192.168.10.64/26 - NOT to 192.168.10.0/26 as intended.

2. Network Address, Broadcast Address and Valid Host Range (Intended Subnet 192.168.10.0/26)

Subnet	Network Address	Broadcast Address	Valid Host Range
192.168.10.0/26 (intended)	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
192.168.10.64/26 (actual, wrong)	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Validity Check of Configured IP and Gateway

- IP 192.168.10.65: INVALID for the intended subnet 192.168.10.0/26 - it actually lies in the neighboring subnet 192.168.10.64/26.
- Gateway 192.168.10.1: VALID for subnet 192.168.10.0/26 (it is the first usable host address of that subnet).

4. Configuration Errors Identified

- The workstation's IP address (192.168.10.65) and its gateway (192.168.10.1) lie in two different subnets.
- Because the host and gateway are not on the same subnet, the workstation cannot reach the gateway or any device in 192.168.10.0/26.

5. Optimized (Corrected) Configuration

- Assign an IP address within the valid range of 192.168.10.0/26, e.g. 192.168.10.2 (avoiding .1, which is reserved for the gateway).
- Subnet Mask: 255.255.255.192 (/26) - unchanged, it was already correct.
- Default Gateway: 192.168.10.1 - unchanged, keep as is.

6. Total Usable Host Addresses After Correction

For a /26 subnet: $2^6 - 2 = 62$ usable host addresses.

Answer 2: IPv6 Configuration Debugging (Device A & Device B)

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 3: Optimizing VLSM Subnet Allocation (192.168.1.0/24)

The /24 block was subnetted using /26, /27 and /28 masks. Using the addresses in the order allocated:

1 & 2. Network Address, Broadcast Address, Host Range, Usable & Unused Addresses

Subnet (as allocated)	Network Addr.	Broadcast Addr.	Valid Host Range	Usable Hosts
192.168.1.0/26	192.168.1.0	192.168.1.63	192.168.1.1 - 192.168.1.62	62
192.168.1.64/27	192.168.1.64	192.168.1.95	192.168.1.65 - 192.168.1.94	30
192.168.1.96/28	192.168.1.96	192.168.1.111	192.168.1.97 - 192.168.1.110	14

Unused block remaining after these three allocations: 192.168.1.112 - 192.168.1.255 (144 addresses, over half of the original /24 left unassigned).

3. Inefficient Subnet Allocations Identified

- Fixed subnet sizes (/26, /27, /28) were assigned without matching actual department host counts, so some departments run out of addresses while a large unused block (144 addresses) remains idle.
- A department needing more than 30 or 14 hosts, but placed in a /27 or /28 subnet, will face address shortage even though free space exists elsewhere in the /24 block.

4. Debugging the Addressing Plan

The root cause is that subnet sizes were chosen arbitrarily instead of being derived from actual/projected host requirements per department (classic error in non-VLSM-aware planning).

5. Recommended Optimized VLSM Scheme

Re-allocate subnets strictly according to actual host requirements, largest first, to minimize wastage. Example VLSM redesign:

Department (example need)	Subnet Assigned	Usable Hosts	Address Range
Dept 1 (~50 hosts)	192.168.1.0/26	62	192.168.1.1 - 192.168.1.62
Dept 2 (~25 hosts)	192.168.1.64/27	30	192.168.1.65 - 192.168.1.94
Dept 3 (~10 hosts)	192.168.1.96/28	14	192.168.1.97 - 192.168.1.110
Dept 4 (~5 hosts)	192.168.1.112/29	6	192.168.1.113 - 192.168.1.118

Remaining block 192.168.1.120 - 192.168.1.255 (136 addresses) is reserved for future departmental growth, giving a much more efficient utilization of the /24 block than the original fixed /26-/27-/28 plan.

Answer 4: IPv6 Configuration Debugging (Duplicate of Question 2)

Question 4 repeats Question 2 verbatim (same Device A and Device B addresses); the analysis and answers are identical to Answer 2 above.

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, 128 - 64 = 64 bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 5: Routing Table Debugging

Routing table: 192.168.1.0/24 -> 10.0.0.2, 192.168.2.0/24 -> 10.0.0.3, Default Route -> 10.0.0.1.

Reported issue: packets destined for 192.168.2.100 are not reaching the intended network.

1. Matching Routing Entry (Longest Prefix Match)

192.168.2.100 falls within 192.168.2.0 - 192.168.2.255, so it matches the entry 192.168.2.0/24 (the only and therefore longest matching prefix).

2. Next-Hop Router Selected

Next hop = 10.0.0.3, as per the matching routing table entry for 192.168.2.0/24.

3. Verification of Destination Subnet

192.168.2.100 is a valid host address within 192.168.2.0/24 (valid host range 192.168.2.1 - 192.168.2.254), so the destination address itself is correctly within the intended subnet.

4. Routing Configuration Errors Identified

The routing table entry itself is correct (192.168.2.0/24 via 10.0.0.3), so the fault is NOT a missing or wrong route entry. The likely causes of failure are operational, such as:

- Next-hop 10.0.0.3 is unreachable, down, or has no return path back to the source network
- Missing reciprocal/return route on the 10.0.0.3 router pointing back to the source subnet
- Interface connected to 10.0.0.3 is administratively down or misconfigured
- An ACL or firewall rule is silently dropping packets destined for 192.168.2.0/24

5. Optimizing the Routing Table

- Verify Layer-2/Layer-3 reachability of next-hop 10.0.0.3 (ping/traceroute) and correct any interface/link fault.
- Add a floating static route or redundant path as backup in case the primary next hop fails.
- Where the topology allows, replace static routes with a dynamic routing protocol (e.g. OSPF) so failed paths are detected and rerouted automatically.
- Confirm the return route exists on the 10.0.0.3 router so response packets are not dropped.

6. Route Used When No Matching Destination Exists

If a destination does not match 192.168.1.0/24 or 192.168.2.0/24, the router forwards it via the Default Route, next hop 10.0.0.1.

Answer 1: Workstation Configuration Debugging (192.168.10.65 /26)

Configured: IP 192.168.10.65, Subnet Mask 255.255.255.192 (/26), Gateway 192.168.10.1. Target subnet: 192.168.10.0/26.

1. Subnet to Which the Configured IP Belongs

192.168.10.65 with a /26 mask (block size 64) actually falls in the range 192.168.10.64 - 192.168.10.127, i.e. it belongs to subnet 192.168.10.64/26 - NOT to 192.168.10.0/26 as intended.

2. Network Address, Broadcast Address and Valid Host Range (Intended Subnet 192.168.10.0/26)

Subnet	Network Address	Broadcast Address	Valid Host Range
192.168.10.0/26 (intended)	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
192.168.10.64/26 (actual, wrong)	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Validity Check of Configured IP and Gateway

- IP 192.168.10.65: INVALID for the intended subnet 192.168.10.0/26 - it actually lies in the neighboring subnet 192.168.10.64/26.
- Gateway 192.168.10.1: VALID for subnet 192.168.10.0/26 (it is the first usable host address of that subnet).

4. Configuration Errors Identified

- The workstation's IP address (192.168.10.65) and its gateway (192.168.10.1) lie in two different subnets.
- Because the host and gateway are not on the same subnet, the workstation cannot reach the gateway or any device in 192.168.10.0/26.

5. Optimized (Corrected) Configuration

- Assign an IP address within the valid range of 192.168.10.0/26, e.g. 192.168.10.2 (avoiding .1, which is reserved for the gateway).
- Subnet Mask: 255.255.255.192 (/26) - unchanged, it was already correct.
- Default Gateway: 192.168.10.1 - unchanged, keep as is.

6. Total Usable Host Addresses After Correction

For a /26 subnet: $2^6 - 2 = 62$ usable host addresses.

Answer 2: IPv6 Configuration Debugging (Device A & Device B)

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^64 = 18,446,744,073,709,551,616$

Answer 3: Optimizing VLSM Subnet Allocation (192.168.1.0/24)

The /24 block was subnetted using /26, /27 and /28 masks. Using the addresses in the order allocated:

1 & 2. Network Address, Broadcast Address, Host Range, Usable & Unused Addresses

Subnet (as allocated)	Network Addr.	Broadcast Addr.	Valid Host Range	Usable Hosts
192.168.1.0/26	192.168.1.0	192.168.1.63	192.168.1.1 - 192.168.1.62	62

192.168.1.64/27	192.168.1.64	192.168.1.95	192.168.1.65 - 192.168.1.94	30
192.168.1.96/28	192.168.1.96	192.168.1.111	192.168.1.97 - 192.168.1.110	14

Unused block remaining after these three allocations: 192.168.1.112 - 192.168.1.255 (144 addresses, over half of the original /24 left unassigned).

3. Inefficient Subnet Allocations Identified

- Fixed subnet sizes (/26, /27, /28) were assigned without matching actual department host counts, so some departments run out of addresses while a large unused block (144 addresses) remains idle.
- A department needing more than 30 or 14 hosts, but placed in a /27 or /28 subnet, will face address shortage even though free space exists elsewhere in the /24 block.

4. Debugging the Addressing Plan

The root cause is that subnet sizes were chosen arbitrarily instead of being derived from actual/projected host requirements per department (classic error in non-VLSM-aware planning).

5. Recommended Optimized VLSM Scheme

Re-allocate subnets strictly according to actual host requirements, largest first, to minimize wastage. Example VLSM redesign:

Department (example need)	Subnet Assigned	Usable Hosts	Address Range
Dept 1 (~50 hosts)	192.168.1.0/26	62	192.168.1.1 - 192.168.1.62
Dept 2 (~25 hosts)	192.168.1.64/27	30	192.168.1.65 - 192.168.1.94
Dept 3 (~10 hosts)	192.168.1.96/28	14	192.168.1.97 - 192.168.1.110
Dept 4 (~5 hosts)	192.168.1.112/29	6	192.168.1.113 - 192.168.1.118

Remaining block 192.168.1.120 - 192.168.1.255 (136 addresses) is reserved for future departmental growth, giving a much more efficient utilization of the /24 block than the original fixed /26-/27-/28 plan.

Answer 4: IPv6 Configuration Debugging (Duplicate of Question 2)

Question 4 repeats Question 2 verbatim (same Device A and Device B addresses); the analysis and answers are identical to Answer 2 above.

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link

- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 5: Routing Table Debugging

Routing table: 192.168.1.0/24 -> 10.0.0.2, 192.168.2.0/24 -> 10.0.0.3, Default Route -> 10.0.0.1.

Reported issue: packets destined for 192.168.2.100 are not reaching the intended network.

1. Matching Routing Entry (Longest Prefix Match)

192.168.2.100 falls within 192.168.2.0 - 192.168.2.255, so it matches the entry 192.168.2.0/24 (the only and therefore longest matching prefix).

2. Next-Hop Router Selected

Next hop = 10.0.0.3, as per the matching routing table entry for 192.168.2.0/24.

3. Verification of Destination Subnet

192.168.2.100 is a valid host address within 192.168.2.0/24 (valid host range 192.168.2.1 - 192.168.2.254), so the destination address itself is correctly within the intended subnet.

4. Routing Configuration Errors Identified

The routing table entry itself is correct (192.168.2.0/24 via 10.0.0.3), so the fault is NOT a missing or wrong route entry. The likely causes of failure are operational, such as:

- Next-hop 10.0.0.3 is unreachable, down, or has no return path back to the source network
- Missing reciprocal/return route on the 10.0.0.3 router pointing back to the source subnet
- Interface connected to 10.0.0.3 is administratively down or misconfigured
- An ACL or firewall rule is silently dropping packets destined for 192.168.2.0/24

5. Optimizing the Routing Table

- Verify Layer-2/Layer-3 reachability of next-hop 10.0.0.3 (ping/traceroute) and correct any interface/link fault.
- Add a floating static route or redundant path as backup in case the primary next hop fails.
- Where the topology allows, replace static routes with a dynamic routing protocol (e.g. OSPF) so failed paths are detected and rerouted automatically.
- Confirm the return route exists on the 10.0.0.3 router so response packets are not dropped.

6. Route Used When No Matching Destination Exists

If a destination does not match 192.168.1.0/24 or 192.168.2.0/24, the router forwards it via the Default Route, next hop 10.0.0.1.

Answer 1: Workstation Configuration Debugging (192.168.10.65 /26)

Configured: IP 192.168.10.65, Subnet Mask 255.255.255.192 (/26), Gateway 192.168.10.1. Target subnet: 192.168.10.0/26.

1. Subnet to Which the Configured IP Belongs

192.168.10.65 with a /26 mask (block size 64) actually falls in the range 192.168.10.64 - 192.168.10.127, i.e. it belongs to subnet 192.168.10.64/26 - NOT to 192.168.10.0/26 as intended.

2. Network Address, Broadcast Address and Valid Host Range (Intended Subnet 192.168.10.0/26)

Subnet	Network Address	Broadcast Address	Valid Host Range
192.168.10.0/26 (intended)	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
192.168.10.64/26 (actual, wrong)	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Validity Check of Configured IP and Gateway

- IP 192.168.10.65: INVALID for the intended subnet 192.168.10.0/26 - it actually lies in the neighboring subnet 192.168.10.64/26.
- Gateway 192.168.10.1: VALID for subnet 192.168.10.0/26 (it is the first usable host address of that subnet).

4. Configuration Errors Identified

- The workstation's IP address (192.168.10.65) and its gateway (192.168.10.1) lie in two different subnets.
- Because the host and gateway are not on the same subnet, the workstation cannot reach the gateway or any device in 192.168.10.0/26.

5. Optimized (Corrected) Configuration

- Assign an IP address within the valid range of 192.168.10.0/26, e.g. 192.168.10.2 (avoiding .1, which is reserved for the gateway).
- Subnet Mask: 255.255.255.192 (/26) - unchanged, it was already correct.
- Default Gateway: 192.168.10.1 - unchanged, keep as is.

6. Total Usable Host Addresses After Correction

For a /26 subnet: $2^6 - 2 = 62$ usable host addresses.

Answer 2: IPv6 Configuration Debugging (Device A & Device B)

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 3: Optimizing VLSM Subnet Allocation (192.168.1.0/24)

The /24 block was subnetted using /26, /27 and /28 masks. Using the addresses in the order allocated:

1 & 2. Network Address, Broadcast Address, Host Range, Usable & Unused Addresses

Subnet (as allocated)	Network Addr.	Broadcast Addr.	Valid Host Range	Usable Hosts
192.168.1.0/26	192.168.1.0	192.168.1.63	192.168.1.1 - 192.168.1.62	62
192.168.1.64/27	192.168.1.64	192.168.1.95	192.168.1.65 - 192.168.1.94	30
192.168.1.96/28	192.168.1.96	192.168.1.111	192.168.1.97 - 192.168.1.110	14

Unused block remaining after these three allocations: 192.168.1.112 - 192.168.1.255 (144 addresses, over half of the original /24 left unassigned).

3. Inefficient Subnet Allocations Identified

- Fixed subnet sizes (/26, /27, /28) were assigned without matching actual department host counts, so some departments run out of addresses while a large unused block (144 addresses) remains idle.
- A department needing more than 30 or 14 hosts, but placed in a /27 or /28 subnet, will face address shortage even though free space exists elsewhere in the /24 block.

4. Debugging the Addressing Plan

The root cause is that subnet sizes were chosen arbitrarily instead of being derived from actual/projected host requirements per department (classic error in non-VLSM-aware planning).

5. Recommended Optimized VLSM Scheme

Re-allocate subnets strictly according to actual host requirements, largest first, to minimize wastage. Example VLSM redesign:

Department (example need)	Subnet Assigned	Usable Hosts	Address Range
Dept 1 (~50 hosts)	192.168.1.0/26	62	192.168.1.1 - 192.168.1.62
Dept 2 (~25 hosts)	192.168.1.64/27	30	192.168.1.65 - 192.168.1.94

Dept 3 (~10 hosts)	192.168.1.96/28	14	192.168.1.97 - 192.168.1.110
Dept 4 (~5 hosts)	192.168.1.112/29	6	192.168.1.113 - 192.168.1.118

Remaining block 192.168.1.120 - 192.168.1.255 (136 addresses) is reserved for future departmental growth, giving a much more efficient utilization of the /24 block than the original fixed /26-/27-/28 plan.

Answer 4: IPv6 Configuration Debugging (Duplicate of Question 2)

Question 4 repeats Question 2 verbatim (same Device A and Device B addresses); the analysis and answers are identical to Answer 2 above.

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
- Device B prefix (first 64 bits): 2001:0db8:0000:0000::/64

3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

4. Addressing / Prefix Mismatch Identified

There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

- Duplicate Address Detection (DAD) conflict or an interface ID typo
- Incorrect or missing default gateway / router advertisement (RA) on the link
- A firewall, ACL, or switch port (VLAN) blocking traffic between the two hosts
- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
- Device B: 2001:db8::20/64

Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, 128 - 64 = 64 bits remain for host addressing:

Total available addresses = $2^{64} = 18,446,744,073,709,551,616$

Answer 5: Routing Table Debugging

Routing table: 192.168.1.0/24 -> 10.0.0.2, 192.168.2.0/24 -> 10.0.0.3, Default Route -> 10.0.0.1.

Reported issue: packets destined for 192.168.2.100 are not reaching the intended network.

1. Matching Routing Entry (Longest Prefix Match)

192.168.2.100 falls within 192.168.2.0 - 192.168.2.255, so it matches the entry 192.168.2.0/24 (the only and therefore longest matching prefix).

2. Next-Hop Router Selected

Next hop = 10.0.0.3, as per the matching routing table entry for 192.168.2.0/24.

3. Verification of Destination Subnet

192.168.2.100 is a valid host address within 192.168.2.0/24 (valid host range 192.168.2.1 - 192.168.2.254), so the destination address itself is correctly within the intended subnet.

4. Routing Configuration Errors Identified

The routing table entry itself is correct (192.168.2.0/24 via 10.0.0.3), so the fault is NOT a missing or wrong route entry. The likely causes of failure are operational, such as:

- Next-hop 10.0.0.3 is unreachable, down, or has no return path back to the source network
- Missing reciprocal/return route on the 10.0.0.3 router pointing back to the source subnet
- Interface connected to 10.0.0.3 is administratively down or misconfigured
- An ACL or firewall rule is silently dropping packets destined for 192.168.2.0/24

5. Optimizing the Routing Table

- Verify Layer-2/Layer-3 reachability of next-hop 10.0.0.3 (ping/traceroute) and correct any interface/link fault.
- Add a floating static route or redundant path as backup in case the primary next hop fails.
- Where the topology allows, replace static routes with a dynamic routing protocol (e.g. OSPF) so failed paths are detected and rerouted automatically.
- Confirm the return route exists on the 10.0.0.3 router so response packets are not dropped.

6. Route Used When No Matching Destination Exists

If a destination does not match 192.168.1.0/24 or 192.168.2.0/24, the router forwards it via the Default Route, next hop 10.0.0.1.

Answer 1: Workstation Configuration Debugging (192.168.10.65 /26)

Configured: IP 192.168.10.65, Subnet Mask 255.255.255.192 (/26), Gateway 192.168.10.1. Target subnet: 192.168.10.0/26.

1. Subnet to Which the Configured IP Belongs

192.168.10.65 with a /26 mask (block size 64) actually falls in the range 192.168.10.64 - 192.168.10.127, i.e. it belongs to subnet 192.168.10.64/26 - NOT to 192.168.10.0/26 as intended.

2. Network Address, Broadcast Address and Valid Host Range (Intended Subnet 192.168.10.0/26)

Subnet	Network Address	Broadcast Address	Valid Host Range
192.168.10.0/26 (intended)	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
192.168.10.64/26 (actual, wrong)	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Validity Check of Configured IP and Gateway

- IP 192.168.10.65: INVALID for the intended subnet 192.168.10.0/26 - it actually lies in the neighboring subnet 192.168.10.64/26.
- Gateway 192.168.10.1: VALID for subnet 192.168.10.0/26 (it is the first usable host address of that subnet).

4. Configuration Errors Identified

- The workstation's IP address (192.168.10.65) and its gateway (192.168.10.1) lie in two different subnets.
- Because the host and gateway are not on the same subnet, the workstation cannot reach the gateway or any device in 192.168.10.0/26.

5. Optimized (Corrected) Configuration

- Assign an IP address within the valid range of 192.168.10.0/26, e.g. 192.168.10.2 (avoiding .1, which is reserved for the gateway).
- Subnet Mask: 255.255.255.192 (/26) - unchanged, it was already correct.
- Default Gateway: 192.168.10.1 - unchanged, keep as is.

6. Total Usable Host Addresses After Correction

For a /26 subnet: $2^6 - 2 = 62$ usable host addresses.

Answer 2: IPv6 Configuration Debugging (Device A & Device B)

1. Expanded (Complete) Notation

- Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix of Each Device (/64)

- Device A prefix (first 64 bits): 2001:0db8:0000:0000::/64
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3. Verification - Same Subnet?

Both Device A and Device B expand to the identical /64 network prefix 2001:db8:0000:0000::/64. Therefore, based on the prefix, both devices DO belong to the same subnet.

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There is NO prefix mismatch - both devices share the same /64 network. Since the subnet portion is correct, the communication failure must be caused by a different fault, such as:

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- Neighbor Discovery Protocol (NDP) cache issue preventing address resolution

5. Optimized IPv6 Configuration

Keep both devices on the common prefix 2001:db8:0000:0000::/64 but assign clearly distinct, non-conflicting interface IDs, e.g.:

- Device A: 2001:db8::10/64
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Verify router advertisement / default gateway configuration and confirm no duplicate address or link-layer (VLAN/ACL) blocking exists.

6. Available Host Addresses in the Corrected Subnet

With a /64 prefix, $128 - 64 = 64$ bits remain for host addressing:

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Answer 3: Optimizing VLSM Subnet Allocation (192.168.1.0/24)

The /24 block was subnetted using /26, /27 and /28 masks. Using the addresses in the order allocated:

1 & 2. Network Address, Broadcast Address, Host Range, Usable & Unused Addresses

Subnet (as allocated)	Network Addr.	Broadcast Addr.	Valid Host Range	Usable Hosts
192.168.1.0/26	192.168.1.0	192.168.1.63	192.168.1.1 - 192.168.1.62	62
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Unused block remaining after these three allocations: 192.168.1.112 - 192.168.1.255 (144 addresses, over half of the original /24 left unassigned).

3. Inefficient Subnet Allocations Identified

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Remaining block 192.168.1.120 - 192.168.1.255 (136 addresses) is reserved for future departmental growth, giving a much more efficient utilization of the /24 block than the original fixed /26-/27-/28 plan.

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Question 4 repeats Question 2 verbatim (same Device A and Device B addresses); the analysis and answers are identical to Answer 2 above.

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2. Next-Hop Router Selected

Next hop = 10.0.0.3, as per the matching routing table entry for 192.168.2.0/24.

3. Verification of Destination Subnet

192.168.2.100 is a valid host address within 192.168.2.0/24 (valid host range 192.168.2.1 - 192.168.2.254), so the destination address itself is correctly within the intended subnet.

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